

Day 17 — MISP: Deployment, IOC Management & TLP

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Objective

Deploy MISP (Malware Information Sharing Platform) via Docker, create 3 realistic threat intelligence events covering different incident types with correctly applied TLP v2.0 classification, enable 3 external threat intelligence feeds, and produce a one-page TLP v2.0 Quick Reference card for MSSP use.

Environment

Deployed on Ubuntu Server (192.168.56.104), the same host used for Wazuh throughout Weeks 1-2. Prior to deployment, disk space was checked (`df -h`) and Docker's unused image cache was cleaned (`docker system prune -a`), reclaiming approximately 6.5GB from old, unused ELK Stack images — Wazuh's active containers were unaffected by this cleanup.

Deployment

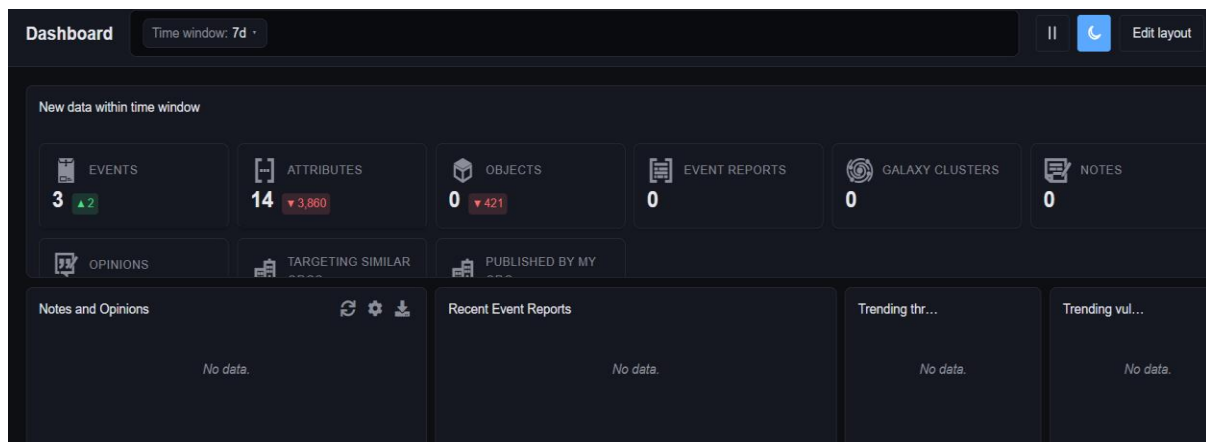
MISP was cloned and configured via the official misp-docker repository:

```
git clone https://github.com/MISP/misp-docker cd misp-docker cp template.env .env
```

The `.env` file was edited to set admin credentials (email, organisation name, password) and the base URL. Ports 80 and 443 were found to already be in use by nginx and Wazuh's dashboard respectively, so MISP was configured to use ports 8080/8443 instead, with the base URL updated to match. MISP was then started:

```
docker compose up -d
```

All 5 containers (misp-core, db, misp-modules, redis, mail) started successfully. The misp-core container briefly showed a Docker "unhealthy" status despite its logs confirming "MISP is now live. Users can now log in" — this was treated as a known, harmless health-check quirk rather than a genuine fault, and was confirmed correct once the web interface loaded successfully at <https://192.168.56.104:8443>.



Event 1 — Phishing Campaign (TLP:RED)

Created an event titled "Phishing Campaign Targeting Finance Team - July 2026", Threat Level High, Analysis Ongoing, Distribution "Your organisation only". The TLP taxonomy was found disabled by default in this MISP instance (Administration > Taxonomies showed tlp with 0/8 active tags) and was enabled before tagging could proceed. Tagged tlp:red, reflecting that this represents a single, unresolved client-specific incident not to be shared further.

4 attributes were added:

- email-src: security-alert@paypa1-verify.com (Payload delivery) — spoofed phishing sender address
- domain: paypa1-verify.com (Payload delivery) — phishing landing page domain
- ip-dst: 185.220.101.47 (Network activity) — IP hosting the phishing landing page
- sha256: (Payload delivery) — hash of the malicious email attachment

Phishing Campaign Targeting Finance Team - July 2026

Event ID	1
UUID	e14ed1ff-2102-4a65-984a-944bd278ca46   
Creator org	CyberShelter
Owner org	CyberShelter
Creator user	admin@cybershelter.local
Protected Event (experimental) 	 Event is in unprotected mode.  Switch to protected mode
Tags	 tlp:red  x  +  +
Date	2026-07-29
Threat Level	 High
Analysis	Ongoing
Distribution	Your organisation only  
Published	Yes First published at: 2026-07-29 13:53:19 Last published at: 2026-07-29 13:53:19
#Attributes	4 (0 Objects)

     Scope toggle   Deleted  Decay score  Context  Related Tags  Filtering tool  Expand all Objects  Collapse all Attributes									
☐	Date 	Context	Category	Type	Value	Tags	Galaxies	Comment	Correlate
☐	2026-07-29*	51b...cc7 	Payload delivery	sha256	44d88612fea8a8f36de82e1278abb02fa3d2a1c9e5b7d4f6a8c0e2f4a6b8c1d3	  	  	Hash of malicious attachment (fake invoice PDF)	☒
☐	2026-07-29*	e7c...3f2 	Network activity	ip-dst	185.220.101.47	  	  	IP hosting the phishing landing page	☒
☐	2026-07-29*	078...8f0 	Payload delivery	domain	paypa1-verify.com	  	  		☒
☐	2026-07-29*	ec0...a2e 	Payload delivery	email-src	security-alert@paypa1-verify.com 	  	  	Spooled sender address used in phishing emails	☒

Event 2 — Ransomware Campaign (TLP:AMBER)

Created an event titled "Ransomware Campaign - C2 Infrastructure Analysis July 2026", Threat Level High, Analysis Ongoing, Distribution "This community only". Tagged tlp:amber, reflecting internal SOC/client-facing analysis not intended for wider distribution.

5 attributes were added:

- ip-dst x2 (Network activity) — primary and secondary C2 servers
- sha256 (Payload delivery) — ransomware encryptor binary
- md5 (Payload delivery) — initial dropper executable
- text (Artifacts dropped) — mutex created by the ransomware to prevent simultaneous re-infection (MISP has no dedicated "mutex" attribute type; the text type with a descriptive comment is the standard workaround)

Ransomware Campaign - C2 Infrastructure Analysis

July 2026

Event ID	2
UUID	c19e6d23-cd0f-449d-9f3e-1ebdb90da2a6   
Creator org	CyberShelter
Owner org	CyberShelter
Creator user	admin@cybershelter.local
Protected Event (experimental) 	 Event is in unprotected mode.  Switch to protected mode
Tags	 tlp:amber   
Date	2026-07-29
Threat Level	 High
Analysis	Ongoing
Distribution	This community only   
Published	 Yes First published at: 2026-07-29 13:54:51 Last published at: 2026-07-29 13:54:51
#Attributes	5 (0 Objects)

☐ Date ↑	Context	Category	Type	Value	Tags	Galaxies	Comment	Correlate
☐ 2026-07-29*	468...c34	Artifacts dropped	text	Global\\RansomLock_8842X			Mutex created by ransomware to prevent multiple simultaneous infections	☒
☐ 2026-07-29*	0b8...92a	Payload delivery	md5	d41d8cd98f00b204e9800998ecf8427e			MD5 of initial dropper executable	☒
☐ 2026-07-29*	918...8f0	Payload delivery	sha256	9f86d081884c7d659a2feaa0c55ad015a3bf4f1b2b0b822cd15d6c15b0f00a08			SHA256 of ransomware encryptor binary	☒
☐ 2026-07-29*	f78...c6f	Network activity	ip-dst	45.153.184.19			Secondary C2 fallback server	☒
☐ 2026-07-29*	6e6...af5	Network activity	ip-dst	192.36.57.192			C2 server for ransomware payload	☒

Event 3 — Credential Stuffing Campaign (TLP:GREEN)

Created an event titled "Credential Stuffing Campaign - Targeted Login Attempts July 2026", Threat Level Medium, Analysis Ongoing, Distribution "Connected communities". Tagged tlp:green, reflecting that this attack-pattern data is genuinely useful to share more broadly with partner MSSPs or an industry group. Domain attributes were placed under the "Network activity" category rather than "Targeting data", since the latter's available types (target-user, target-email, target-org, etc.) do not include a "domain" type.

5 attributes were added:

- ip-src x3 (Network activity) — source IPs observed in credential stuffing attempts
- domain x2 (Network activity) — client login/webmail portals targeted

Credential Stuffing Campaign - Targeted Login

Attempts July 2026

Event ID	3
UUID	95e6f0ad-22c6-47c0-8d77-c19e871cb262
Creator org	CyberShelter
Owner org	CyberShelter
Creator user	admin@cybershelter.local
Protected Event (experimental)	Event is in unprotected mode. Switch to protected mode
Tags	tip:green + +
Date	2026-07-29
Threat Level	Medium
Analysis	Ongoing
Distribution	Connected communities
Published	Yes First published at: 2026-07-29 13:54:23 Last published at: 2026-07-29 13:54:23

#Attributes		5 (0 Objects)							
☐ Date ↑	Context	Category	Type	Value	Tags	Galaxies	Comment	Correlate	
☐ 2026-07-29*	f92...a80	Network activity	domain	webmail.cybershelter-client2.com	+ +	+ +	Client webmail portal targeted by credential stuffing	☒	
☐ 2026-07-29*	c95...1b1	Network activity	domain	portal.cybershelter-client1.com	+ +	+ +	Client login portal targeted by credential stuffing	☒	
☐ 2026-07-29*	d45...fd2	Network activity	ip-src	45.148.10.183	+ +	+ +	Source IP observed in credential stuffing attempts	☒	
☐ 2026-07-29*	55a...ea7	Network activity	ip-src	194.31.98.6	+ +	+ +	Source IP observed in credential stuffing attempts	☒	
☐ 2026-07-29*	254...6cb	Network activity	ip-src	103.224.182.245	+ +	+ +	Source IP observed in credential stuffing attempts	☒	

All 3 events were published

Publishing was completed for full workflow correctness, though since this is a standalone lab MISP instance with no configured sync servers, published events are not actually transmitted to any external organisation.

Threat Feed Configuration

CIRCL OSINT Feed was present in MISP's default feed list and was enabled directly. MalwareBazaar and Abuse.ch URLhaus were not present by default and were added manually via "Add Feed", using the official MISP-format feed URLs (bazaar.abuse.ch/downloads/misp/ and urlhaus.abuse.ch/downloads/misp/ respectively) rather than their CSV equivalents, to benefit from MISP's full native format.

"Fetch and store all feed data" was run to populate the feeds. During this operation, resource contention was observed: the misp-core and db containers consumed sustained high CPU while processing years of historical feed events, and this coincided with all 3 Wazuh containers being terminated by the Linux OOM (out-of-memory) killer (confirmed via exit codes 137 and 143 on wazuh-dashboard and wazuh-indexer respectively). Wazuh was left stopped for the remainder of the session, as it was not required for this task and its data/images remained intact — only the running processes were affected.

The feed fetch completed successfully, confirmed via MISP's Jobs queue (Administration > Jobs) showing all publish_event jobs as "Completed", and via the Events list, which showed 240+ new events beyond the original 3 — including entries explicitly named "MalwareBazaar malware samples for [date]", "URLhaus IOCs for [date]", and genuine historical CIRCL OSINT reports (e.g., "OSINT - Locky: the encryptor taking the world by storm") — confirming all 3 required feeds were genuinely active and populated with real data, not just toggled on.

Feeds

Generate feed lookup caches or fetch feed data (enabled feeds only)

Load default feed metadata

Cache all feeds

Cache freetext/CSV feeds

Cache MISP feeds

Fetch and store all feed data

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Default feeds Custom feeds All feeds Enabled feeds										Enter value to search		Filter			
☐	ID	Enabled	Caching	Name	Format	Provider	Org	Source	URL	Headers	Target	Publish	Delta	Override	Distribution
☐	1	✓	✗	CIRCL OSINT Feed	misp	CIRCL		network	https://www.circl.lu/doc/misp/feed-osint			✗	✗	✗	All communities
☐	3	✓	✗	MalwareBazaar	misp	abuse.ch		network	https://bazaar.abuse.ch/downloads/misp/			✗	✗	✗	All communities
☐	4	✓	✗	URLhaus	misp	abuse.ch		network	https://urlhaus.abuse.ch/downloads/misp/			✗	✗	✗	All communities

Events

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2

3

4

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next »

My EventsOrg Events											Enter value to search		Event info	Filter
☐	Creator org	Owner org	ID	Clusters	#Attr.	#Corr.	Creator user	Date	Last modified at ↑	Published at	Info	Distribution		
☐	✓ CyberShelter	CyberShelter	3		5		admin@cybershelter.local	2026-07-29	2026-07-29 13:45:12	2026-07-29 13:54:23	Credential Stuffing Campaign - Targeted Login Attempts July 2026	Connected		
☐	✓ CyberShelter	CyberShelter	2		5	1	admin@cybershelter.local	2026-07-29	2026-07-29 13:36:19	2026-07-29 13:54:51	Ransomware Campaign - C2 Infrastructure Analysis July 2026	Community		
☐	✓ CyberShelter	CyberShelter	1		4		admin@cybershelter.local	2026-07-29	2026-07-29 13:21:07	2026-07-29 13:53:19	Phishing Campaign Targeting Finance Team - July 2026	Organisation		
☐	✓ abuse.ch	CyberShelter	263		3176	24	admin@cybershelter.local	2026-07-17	2026-07-18 00:03:02		MalwareBazaar malware samples for 2026-07-17	All		
☐	✓ abuse.ch	CyberShelter	159		2652	34	admin@cybershelter.local	2026-05-17	2026-05-18 00:03:01	2026-07-29 14:26:52	MalwareBazaar malware samples for 2026-05-17	All		
☐	✓ abuse.ch	CyberShelter	27		3921	40	admin@cybershelter.local	2026-04-07	2026-04-08 00:03:01	2026-07-29 14:18:56	MalwareBazaar malware samples for 2026-04-07	All		
☐	✓ abuse.ch	CyberShelter	258		3805	36	admin@cybershelter.local	2026-03-23	2026-03-24 00:03:02	2026-07-29 14:35:40	MalwareBazaar malware samples for 2026-03-23	All		

☐	✓ abuse.ch	CyberShelter	1/3		42/01	31	admin@cybershelter.local	2021-04-29	2021-04-30 00:03:02		MalwareBazaar malware samples for 2021-04-29	All
☐	✓ abuse.ch	CyberShelter	260		3568	34	admin@cybershelter.local	2021-04-22	2021-04-23 00:03:01	2026-07-29 14:36:30	MalwareBazaar malware samples for 2021-04-22	All
☐	✓ abuse.ch	CyberShelter	73		4087	24	admin@cybershelter.local	2021-04-05	2021-04-06 00:03:02	2026-07-29 14:22:31	MalwareBazaar malware samples for 2021-04-05	All
☐	✓ CthulhuSPRL.be	CyberShelter	251		2308	3	admin@cybershelter.local	2015-11-05	2020-08-03 06:40:59	2026-07-29 14:33:53	OSINT Expansion on Systematic cyber attacks against Israeli and Palestinian targets going on for a year by Norman	All
☐	✓ CyberShelter	CyberShelter	88		6		admin@cybershelter.local	2016-04-08	2018-12-12 13:53:11	2026-07-29 14:21:49	OSINT - Locky: the encryptor taking the world by storm	All
☐	✓ CthulhuSPRL.be	CyberShelter	243		7440	8	admin@cybershelter.local	2014-11-20	2018-02-05 07:53:58	2026-07-29 14:34:10	Import of CitizenLab public DB of malware indicators	All
☐	✓ CyberShelter	CyberShelter	147		110	4	admin@cybershelter.local	2016-05-23	2018-01-13 03:00:22	2026-07-29 14:24:43	OSINT - Operation Ke3chang Resurfaces With New TidePool Malware	All
☐	✓ CthulhuSPRL.be	CyberShelter	248	Threat Actor		3	admin@cybershelter.local	2016-05-13	2017-06-22 20:12:20	2026-07-29 14:33:23	OSINT DarkHotel samples at WooYun	All
☐	✓ CthulhuSPRL.be	CyberShelter	247		110	2	admin@cybershelter.local	2016-07-11	2017-06-22 20:12:07	2026-07-29 14:33:23	What's in a server name (on APT28/Sofacy) by ThreatConnect	All
☐	✓ CyberShelter	CyberShelter	218		113	6	admin@cybershelter.local	2016-07-13	2017-05-02 13:26:55	2026-07-29 14:32:13	Malspam 2016-07-13 (Locky campaign 'Profile')	All

Key Concepts Applied

- Event — one incident/campaign container (the 3 created today)
- Attribute — a single IOC within an event (individual IPs, hashes, domains, the email address)
- Object — a structured bundle of related attributes describing one thing; not required for today's simpler attribute sets, but understood as the mechanism MISP uses to group, e.g., a file's hash + filename + size as one cohesive unit
- Tag — a label attached to an event; TLP tags were the primary type used, though MISP also supports tags for frameworks like MITRE ATT&CK
- TLP v2.0 — CLEAR/GREEN/AMBER/RED, with TLP:WHITE formally retired and renamed to TLP:CLEAR since August 2022

Deliverable

- MISP running (Docker) with 3 events, correct TLP tags, and 3 active feeds populated with real data
- TLP_v2_Quick_Reference.docx — one-page card covering all 4 labels and MSSP-context usage guidance

Conclusion

Today marked the first platform-deployment day of Week 3, shifting from Python scripting to standing up and configuring a real threat-intelligence application. Beyond the core task, a genuine infrastructure incident (Wazuh's OOM termination during heavy feed ingestion) was correctly diagnosed via container exit codes rather than assumed, and resolved pragmatically by deprioritising a service not needed for the day's work. The distinction between MISP's Event, Attribute, Object, and Tag concepts, along with the correct, current TLP v2.0 labelling (including the CLEAR rename), are concepts directly transferable to any client MISP deployment going forward.